Classification and Magnitude Estimation of Global and Local Seismic Events Using Conformer and Low-Rank Adaptation Fine-Tuning

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***Abstract*— Classifying seismic events and estimating their magnitude are crucial topics in the study of seismic waves. Due to the disparities between global and local geologic features, models exclusively trained on global data may exhibit suboptimal performance in local contexts. To solve this problem, this letter proposes a method to evaluate the effectiveness of the low-rank adaptation (LoRA) technique in seismic wave research using the convolution-augmented transformer (Conformer). We simplified and modified the Conformer model, reducing the number of parameters by more than 169-fold, and applied the LoRA technique to this model. Experimental results using the Stanford Earthquake Dataset (STEAD) and the Korean Peninsula Earthquake Dataset (KPED) from 2017 to 2018 showed that fine-tuning the model with a significantly reduced number of parameters using the proposed method is suitable for research on seismological applications. Our approach achieved over 99.99% accuracy in seismic event classification for both datasets. Additionally, our model demonstrated a 7% decrease in mean absolute error (MAE) on the STEAD dataset and a 48% decrease on the KPED dataset compared to the state-of-the-art model. Furthermore, the results also indicate that the Conformer is suitable for seismic event classification and magnitude estimation. The model's performance in the seismic event classification task decreased by 0.1%, despite reducing the number of retrain parameters by 59 times. Additionally, in the magnitude estimation task, there was an 89-fold decrease in the number of retrain parameters, yet the performance decreased by 1%.**

***Index Terms*— Attention, Conformer, convolutional neural network (CNN), deep learning, low-rank adaptation (LoRA), magnitude estimation, seismic event classification.**

I. Introduction

THE investigation into seismicity serves as an essential tool for understanding earthquake occurrences, their geographical distribution, and distinctive features, laying the foundation for a comprehensive comprehension of these geological phenomena. Furthermore, seismicity research assesses seismic hazards in different regions, identifies potential earthquake-prone areas, and provides valuable information for disaster preparedness, surveillance [1], and mitigation efforts, playing a crucial role in saving lives and minimizing earthquake damage. In earthquake analysis and seismological research, identifying earthquake events and determining their magnitude are primary parameters of significance. The short-term average/long-term average (STA/LTA) [2] algorithm is widely utilized in seismology for detecting and characterizing seismic signals, particularly earthquakes. It involves calculating the ratio between the STA and LTA of a seismic signal. When the STA/LTA ratio surpasses a certain threshold, it indicates the presence of a notable seismic event, making it an effective tool for earthquake monitoring and analysis. However, the ability to detect seismic events using the STA/LTA process depends on the predefined threshold and is associated with several drawbacks. These limitations include the occurrence of numerous false alarms and diminished performance in the presence of significant noise. In the task of magnitude estimation, the Richter scale or Gutenberg–Richter scale [3] is widely recognized as the most common method for expressing earthquake magnitude, initially developed by Charles Richter in 1935. Since then, various methods for measuring earthquake magnitude have been proposed. For example, Kanamori [4] determined magnitude using the duration of shaking recorded on a seismogram.

Due to its suitability for processing seismic waveforms, deep learning seismology has gained wide utilization and popularity. Convolutional neural network (CNN) [5], [6], [7], [8], [9], [10], [11], [12], [13], recurrent neural network (RNN) [9], [10], [11], [14], and attention algorithm [11], [15] have been utilized in several deep learning models for both seismic event detection/classification and magnitude estimation. For example, Detnet [6] and Yews [7] utilized very deep CNNs and max-pooling layers for seismic event classification. CRED [10] is an earthquake signal detection model that incorporates not only 2-D CNNs but also bidirectional long short-term memory (LSTM) networks [16] in a residual architecture. Kim [12] demonstrated the efficacy of fusing deep CNNs and graph convolutional networks (GCNs) for seismic event classification, utilizing raw waveform data from multiple stations. EQTransformer [11] has achieved remarkable results in earthquake detection using CNNs, LSTMs, and attention mechanisms. Regarding magnitude estimation, MagNet [14] estimates earthquake magnitudes using CNNs and bidirectional LSTMs, while Complex CNN [8] employs CNNs with complex-valued representations of seismograms. DeeperCRNN [9] estimates magnitudes by utilizing a very deep convolutional block, comprising several convolution, normalization, max-pooling, and dropout layers, as well as an embedding network designed for extracting features from the maximum amplitude of seismic waveforms.

However, all of these related works train their models using specific datasets in their studies. Earthquake characteristics

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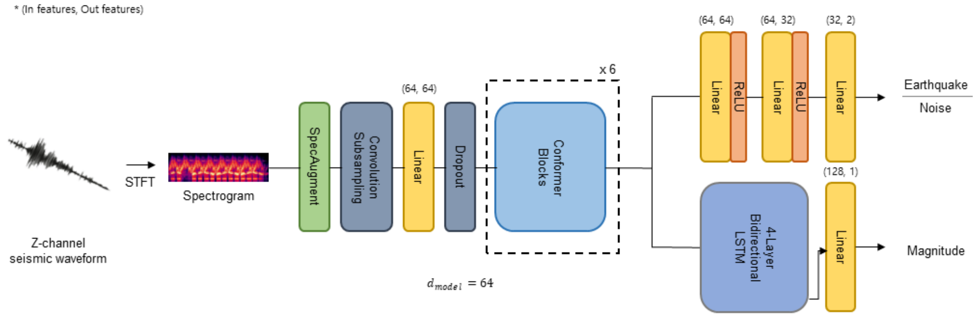


Fig. 1. Proposed architecture for seismic event classification and magnitude estimation. Specaugment was applied to the spectrogram created by applying STFT to the Z-channel waveform, which was then utilized as input. The encoder consists of convolutional subsampling layers, linear layers, dropout layers, and our simplified and modified Conformer architecture. For the seismic event classification task, the decoder comprises a series of linear layers with ReLU activation. In contrast, for the magnitude estimation task, the decoder consists of a four-layer bidirectional LSTM followed by a linear layer.

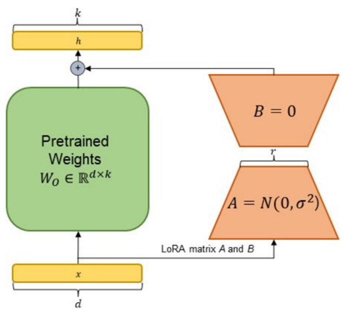


Fig. 2. Detailed structure of LoRA matrix. During the fine-tuning phase, we solely train the low-rank decomposed matrices A and B while maintaining the pretrained weights in the Conformer model.

can vary within each country, differing from region to region. For instance, Korea is not situated along a plate boundary, resulting in a smaller number of earthquakes but a higher occurrence of micro-earthquakes. In contrast, Japan and the western United States, located within the circum-Pacific orogeny belt, experience frequent earthquakes with a higher average intensity. Attenuation and site amplification represent two primary components of magnitude scale relationships, such as M_L, which vary from one region to another. Therefore, a global model is unlikely to perform optimally across all regions. Performing fine-tuning on all parameters to address this challenge can be challenging due to the large number of parameters to be relearned. The low-rank adaptation (LoRA) method [17], presented as a parameter-efficient fine-tuning approach for language models, offers an effective solution to this problem. This approach significantly reduces the number of trainable parameters for subsequent tasks by maintaining the pretrained model weights fixed and integrating a trainable rank decomposition matrix into each layer of the model. If a model is trained on a frequently utilized global dataset and then fine-tuned using the LoRA method with a small number of parameters for a specific dataset, it would be beneficial for analyzing the seismic characteristics and conducting seismic signal research in a particular region.

In this letter, we propose seismic event classification and magnitude estimation models utilizing the Conformer architecture [18]. Additionally, we introduce a fine-tuning approach based on LoRA to adapt classification and magnitude estimation for local events from a large-scale global model. The rationale for selecting the Conformer as the model for seismicity research and as the model to apply the LoRA method is as follows: seismic signals are time-series data containing both local and global features. Therefore, the combination of attention mechanisms and CNNs in seismicity research offers several advantages. It facilitates effective feature extraction using CNNs, enabling the model to capture complex patterns indicative of seismic events. The attention mechanism assists in capturing long-distance relationships in seismic data by utilizing multiple weight matrices for query, key, and value. Furthermore, it allows the model to flexibly assign attention to distinct features or channels based on their relevance to the given task. This adaptability enhances the model's ability to capture pertinent information while disregarding extraneous noise or background signals. From this perspective, the Conformer, which integrates both CNNs for effectively capturing local features and attention modules for effectively capturing global features within the Conformer block, is expected to demonstrate optimal effectiveness in seismicity research.

The contributions of our work are as follows.

1) We propose a novel Conformer-based model for seismic event classification and magnitude estimation.

2) By applying the LoRA method to seismicity research, we efficiently address the challenge of diverse characteristics exhibited by seismic waveforms across different regions.

The subsequent sections of this letter are structured as follows. Section II details the proposed method for seismic event classification and magnitude estimation. Section III presents the experimental process and results to evaluate our work. Section IV provides conclusions drawn from the study.

II. Proposed Method

Fig. 1 illustrates the architecture of the proposed network. The network takes the spectrogram from the Z-component seismogram as input, which can be obtained through the application of the discrete short-time Fourier transform (STFT). Initially, the spectrogram undergoes processing by the convolution subsampling layer. Subsequently, it passes through a linear layer with 64 input features and 64 output features. Dropout with $p = 0.1$ is then applied. Data augmentation is consistently performed during training using SpecAugment [19].

In the encoder architecture, we utilize the Conformer structure [18]. A Conformer block consists of four modules arranged sequentially: a feed-forward module, a self-attention module, a convolution module, and a final feed-forward module. The two feed-forward networks employed are referred to as half-step feed-forward networks, inspired by Macaron-Net [20]. A residual connection [21] is incorporated after each pass through the Conformer block's modules. Layer normalization is applied after completing all module passes and before entering the subsequent Conformer block. Dropout with $p = 0.1$ is applied after every linear layer. Unlike the model presented in the original paper, we modify the model architecture by simplifying it. First, we reduce the number

TABLE I
MAGNITUDE AND NUMBER OF EARTHQUAKES IN THE EXPERIMENTAL DATA. M DENOTES THE MAGNITUDE OF THE EARTHQUAKE

STEAD	$M < 2$	$2 \leq M < 4$	$M \geq 4$	Total	Noise
	758,896	167,233	104,152	1,030,231	235,426
KPED	$M < 2$	$2 \leq M < 4$	$M \geq 4$	Total	Noise
	176,301	129,561	0	305,862	30,000

of layers. While the original model utilized over 16 encoder layers, we employ only six layers. Next, we further reduce the dimension of the model. While the encoder in the original paper had a dimension of 512, we set the encoder dimension to 64. Through this simplification, we achieve a substantial decrease in the number of parameters compared to the original paper's configuration. Our proposed encoder model has only 0.7 M parameters, whereas the original Conformer has 118.8 M parameters.

In seismic event classification, we employ three fully connected (FC) layers for the decoder. The output dimensions of each FC layer are 64, 32, and 2, respectively. Except for the final FC layer, ReLU activation function and dropout with $p = 0.1$ are applied consistently. For the magnitude estimation decoder, we utilize a four-layer bidirectional LSTM with a hidden dimension of 128, which is twice the size of the encoder dimension, and a single FC layer with 128 hidden units. Dropout with $p = 0.1$ is applied before the FC layer.

During the fine-tuning phase, the LoRA technique is employed. The specific methodology of LoRA unfolds as follows: the pre-trained weight matrix, denoted as W_O with dimensions (d, k), serves as the foundation for the anticipated dimensions of the accumulated gradient values, ΔW, which are expected to manifest as (d, k). These are represented as $\Delta W = BA$, utilizing a row-rank parameter r, where B has dimensions of (d, r) and A has dimensions of (r, k). Throughout the training process, W_O remains unchanged through gradient updates, while the focus of the learning mechanism revolves around the manipulation of BA. Fig. 2 illustrates the specific structure of LoRA.

III. EXPERIMENTS

In this letter, we utilized two datasets (Table I). First, we employed the Stanford Earthquake Dataset (STEAD) [22], which comprises 1.03 million three-component seismograms and 200 000 noises associated with earthquakes occurring between 1984 and 2018. Second, we utilized the Korean Peninsula Earthquake Dataset (KPED), consisting of 305 000 three-component seismograms and 30,000 noises recorded between 2017 and 2018. The creation of KPED followed the methodology outlined in [12]. During the period from January 2017 to 2018, seismic data from the Korean Peninsula were collected, and the geographical positions of the seismic stations are depicted in Fig. 3. Three distinct sensor types were employed: HH (broadband seismometer), EL (short-period seismometer), and HG (broadband accelerometer). Each sensor recorded data in three channels: E (west-east), N (north-south), and Z (vertical), with a sampling rate of 100 Hz. To enhance training speed, we exclusively utilized data from the Z-channel.

For the experiments, we utilized the PyTorch framework. We set the learning rate to 1e-4 and the weight decay rate to 1e-5. We employed the Adam optimizer [23] with $\beta_1 = 0.9$ and $\beta_2 = 0.99$. For the loss function, we utilized mean-squared error for seismic event classification and cross-entropy error for magnitude estimation. The experiment was halted when the validation accuracy did not improve

Fig. 3. Seismic station deployment in South Korea.

for 10 consecutive epochs. Regarding dataset partitioning, we divided both datasets into an 85:5:10 proportion for the training, validation, and test datasets, respectively. The training process was completed in 12 h using a single NVIDIA GeForce RTX 2080 Ti GPU.

A. Preprocessing

The observed seismic data may exhibit varying biases depending on each individual observation station. Normalization helps mitigate the influence of biases, outliers, or extreme values, ensuring a more balanced and reliable analysis. By minimizing the impact of these extreme values, the analysis becomes more robust and less prone to skewed results. Hence, we normalized each data sample using the following equation:

$$y[m] = x[m] - \frac{1}{M}\sum_{m=1}^{M} x[m], \quad m = 1, \ldots, M \tag{1}$$

where M represents the total number of sample points in the data.

B. Data Augmentation

For the KPED, which contains both 40- and 65-s waveforms, we applied data augmentation techniques using time-shifting methods. For the 40-s waveforms, we shifted the waveform by 0.5-s intervals in padding to achieve a total length of 60 s, resulting in 20 times data augmentation for seismic waveforms. For the 65-s waveforms, we also cut the data into 60-s segments by changing the padding interval by 0.5 s, resulting in ten times data augmentation.

C. Experimental Results

Initially, in the seismic event classification task, we conducted a comparative analysis between the proposed method and five existing methods: EQTransformer [11], CRED [10], DetNet [6], Yews [7], and STA/LTA [2]. For measurement of the proposed model in seismic event classification, we utilized the following four indices:

$$\text{Accuracy(Acc)} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

$$\text{Precision(Pr)} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

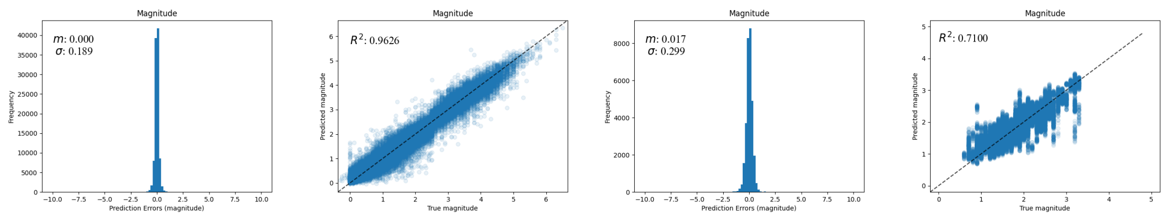
$$\text{Recall(Re)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$F1 - \text{Score}(F1) = \frac{2 \times \text{Precision} \times \text{Recll}}{\text{Precision} + \text{Recall}}$$

TABLE II

RESULTS IN SEISMIC EVENT CLASSIFICATION

		STEAD				KPED			
Models	# of params	Acc[%]	Pr[%]	Re[%]	F1[%]	Acc[%]	Pr[%]	Re[%]	F1[%]
EQTrasnformer [11]	372K	99.03	99.69	99.11	99.40	99.97	99.98	99.97	99.98
CRED [10]	256K	99.63	99.76	99.79	99.77	**100.00**	**100.00**	**100.00**	**100.00**
DetNet [6]	5K	98.27	99.26	98.60	98.93	99.46	99.90	99.37	99.63
Yews [7]	108K	93.73	96.15	96.14	96.15	83.73	89.27	93.12	91.15
STA/LTA [2]	-	94.79	96.52	97.10	96.81	61.86	**100.00**	48.05	64.91
Ours	714K	**99.99**	**99.99**	**99.99**	**99.99**	**100.00**	**100.00**	**100.00**	**100.00**



(a) (b) (c) (d)

Fig. 4. Prediction results achieved by Conformer on the test subset of STEAD and KPED. The error distribution histograms for magnitude in the STEAD and KPED are presented in the first and third columns, respectively. The mean (m) and standard deviation (σ) of the errors are given in both the first and third columns. The comparative analysis of predicted and observed magnitudes in the STEAD and KPED is depicted in the second and fourth columns, respectively. The ideal outcome, denoted by the dotted line, signifies perfect agreement with zero error. The coefficient of determination (R^2) is also included for reference. (a) Error distribution histogram for magnitude in the STEAD. (b) Predicted and observed magnitudes in the STEAD. (c) Error distribution histogram for magnitude in the KPED. (d) Predicted and observed magnitudes in the KPED.

TABLE III

RESULTS IN MAGNITUDE ESTIMATION

		STEAD	KPED
Models	# of params	MAE	MAE
Complex CNN [8]	-	0.26	-
Magnet [14]	387K	0.2647	0.3711
Deeper CRNN [9]	1.84M	0.1377	0.2849
Ours	1.07M	**0.1278**	**0.1925**

TABLE IV

EXPERIMENTAL RESULTS CONDUCTED BY ADJUSTING THE HYPERPARAMETER r OF LoRA

r	# of retrain params	Acc[%]	Pr[%]	Re[%]	F1[%]	MAE
1	12K	**99.90**	**99.99**	**99.92**	**99.95**	**0.2811**
2	25K	99.82	99.91	99.84	99.87	0.2830
4	51K	99.61	99.78	99.59	99.68	0.2813
8	101K	99.82	99.80	**99.92**	99.85	0.2820
16	203K	99.70	99.87	99.81	99.83	**0.2811**

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively. Table II provides a summary comparison between our proposed method and existing models using these terms. The results of all existing methods in Table II were obtained by training and testing under the same experimental conditions as our proposed method. Our experiments encompass both the STEAD dataset and the KPED, thus Table II comprehensively presents the experimental findings for both datasets. The entry "STEAD" in the initial row of Table II denotes the test outcomes derived from the STEAD dataset, while "KPED" signifies the test results obtained from the Korean Peninsula Earthquake Dataset. Table II demonstrates that our proposed method exhibits superior performance in seismic event classification tasks for both datasets compared to other existing methods.

We also performed a comparative analysis between the proposed method and three existing models: Complex CNN [8], Magnet [14], and Deeper CRNN [9], in the task of magnitude estimation. For measurement in magnitude estimation, we utilized mean absolute error (MAE) for performance evaluation. MAE, which quantifies the disparity between the actual values and their corresponding predicted values, is defined by the following equation:

$$\mathrm{MAE} = \frac{1}{M}\sum_{m=1}^{M} ||x_{gt}[m] - x_{\mathrm{pred}}[m]||$$

where M represents the total number of dataset, $x_{gt}[m]$ represents mth ground-truth magnitude, and $x_{\mathrm{pred}}[m]$ represents mth predicted magnitude. Table III presents the reported performance of Complex CNN, while the evaluations of the remaining existing models were obtained through training and testing under identical experimental conditions as our proposed method. Similar to Table II, "STEAD" and "KPED" in the first row of Table III represent the experimental results in STEAD and Korean Peninsula Earthquake Dataset, respectively. Table III shows that the proposed method outperforms existing methods in magnitude estimation. Fig. 4 displays the histograms illustrating the distribution of errors, as well as the comparison between expected and actual magnitude values for both datasets.

To assess the impact of the LoRA method, we implemented the LoRA technique on all linear layers within our proposed model. To determine the low-rank parameter r, we conducted several experiments using different rank numbers. Table IV illustrates that as the rank number increases, the number of parameters requiring retraining also increases, but the performance eventually saturates. Therefore, we opted to set the low-rank parameter r to 1 for both tasks. Table V presents the results of applying the LoRA method along with a comparison with other methods. All test results in Table V are evaluated using KPED's test dataset. Due to LoRA's limitation in combining diverse tasks into batches—each task employs distinct approaches for configuring LoRA parameters—we conducted fine-tuning individually. After evaluating the model on KPED's test dataset using weights trained with STEAD, we observed a significant decrease in performance. However, when we performed fine-tuning of all model weights using KPED, the model exhibited exceptional performance. Nonetheless, this process entailed the retraining of a substantial number of weights. The results obtained using the

TABLE V

RESULTS IN APPLYING LORA METHOD. STEAD WAS UTILIZED AS THE PRETRAIN DATASET AND KPED WAS UTILIZED FOR THE TEST DATASET

Task	Model	Method	# of Retrain params	Acc[%]	Pr[%]	Re[%]	F1[%]	MAE
Seismic Event Classification	Conformer	Without Fine-tuning	0	60.99	86.60	55.43	67.59	-
		Fine-tune all params	714K	100.00	100.00	100.00	100.00	-
		LoRA	12K	99.90	99.99	99.92	99.95	-
Magnitude Estimation	Conformer	Without Fine-tuning	0	-	-	-	-	0.5957
		Fine-tune all params	1.07M	-	-	-	-	0.2758
		LoRA	12K	-	-	-	-	0.2811

TABLE VI

ABLATION STUDY FOR THE CONVOLUTION LAYER AND ATTENTION LAYER IN CONFORMER NETWORK

Model	Acc[%]	Pr[%]	Re[%]	F1[%]
Remove Convolution Layer	97.11	97.45	97.95	97.69
Remove Attention Layer	96.75	97.01	98.24	97.62
Conformer	**99.99**	**99.99**	**99.99**	**99.99**

LoRA method are presented in the third and sixth rows of the table. In the seismic event classification task, the performance achieved when retraining with the LoRA method while retaining the pretrained weights of the model exceeded 99% compared to the result obtained by retraining all weights. Even for the magnitude estimation task, there was only a slight increase in the MAE, by approximately 0.005. In both tasks, the number of weights that needed retraining was about 12K, resulting in a reduction of over 50 times in the number of weights that required retraining. These results address the uncertainty presented in the introduction regarding whether a model pretrained using a specific dataset will perform well on other seismic datasets with different characteristics. Furthermore, these findings demonstrate the effective application of the LoRA method to seismic data with diverse characteristics, even when utilizing a limited number of parameters.

D. Ablation Study

To demonstrate the effectiveness of both the CNN layer and the attention layer in the Conformer block for seismic waveform analysis, we conducted some ablation studies. We trained models with the convolution layer removed from the Conformer block and models with the attention layer removed on the STEAD dataset. As shown in Table VI, the performance decreased further when both the convolution layer and the attention layer were removed. Through this analysis, as explained in the introduction, we can demonstrate that both the CNN layer, effective for analyzing local features, and the attention layer, effective for analyzing global features, contribute significantly to the analysis of seismic waveforms.

IV. CONCLUSION

In this letter, we introduced a streamlined and customized Conformer architecture designed for seismic event classification and magnitude estimation. Furthermore, we demonstrated the efficacy of the LoRA method in seismicity research. Through the experimental analysis conducted on both the STEAD and KPED, which possess distinct characteristics, the Conformer model demonstrated superior performance compared to existing models in both seismic event classification and magnitude estimation tasks. Moreover, the investigation revealed that both the Convolution module and Attention module of the Conformer contribute significantly to the processing of seismic signals. In addition, it was observed that the Seismic Event Classification model and Magnitude Estimation model, trained on different datasets, yielded promising outcomes when applied to additional datasets with minimal parameter training using the LoRA method.

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